

```
# Importing libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error
```

```
# --- NumPy Example --
print("NumPy Array Operations")
arr = np.array([1, 2, 3, 4, 5])
print("Array:", arr)
print("Mean:", np.mean(arr))
print("Standard Deviation:", np.std(arr))
```



```
NumPy Array Operations
Array: [1 2 3 4 5]
Mean: 3.0
Standard Deviation: 1.4142135623730951
```

```
# --- Pandas Example --
print("\nPandas DataFrame Example")
data = {'Name': ['Alice', 'Bob', 'Charlie'], 'Marks': [85, 90, 95]}
df = pd.DataFrame(data)
print(df)
print("Average Marks:", df['Marks'].mean())
```



```
Pandas DataFrame Example
   Name  Marks
0  Alice    85
1   Bob    90
2 Charlie    95
Average Marks: 90.0
```

```
# --- Scikit-Learn Example (Linear Regression) --
print("\nScikit-Learn Linear Regression Example")
# Generating sample data
X = np.array([[1], [2], [3], [4], [5]])
y = np.array([1.2, 1.9, 3.2, 3.9, 5.1])
```



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Scikit-Learn Linear Regression Example
```

```
# Splitting data
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

```
# Creating and training model
model = LinearRegression()
model.fit(X_train, y_train)
```



▼ LinearRegression ⓘ ?

LinearRegression()

```
# Predicting
y_pred = model.predict(X_test)
print("Mean Squared Error:", mean_squared_error(y_test, y_pred))
```



Mean Squared Error: 0.06612244897959182

```
# --- Matplotlib Example ---
print("\nMatplotlib Plot")
plt.scatter(X, y, color='blue', label='Original Data')
plt.plot(X, model.predict(X), color='red', label='Regression Line')
plt.title("Linear Regression Plot")
plt.xlabel("X")
plt.ylabel("y")
plt.legend()
plt.grid(True)
plt.show()
```



Matplotlib Plot

